

KnowSelf: Agentic Knowledgeable Self-Awareness for LLM-Based Agents

Enabling agents to autonomously regulate knowledge utilization through situational self-awareness.

Paper: arxiv.org/abs/2504.03553

Current Agent Training Uses 'Flood Irrigation'

Indiscriminate Knowledge Injection Hurts Efficiency

Existing approaches blindly inject gold trajectories, external feedback, and domain knowledge without considering situational needs. This limits efficiency and leads to pattern collapse.

Trajectory Imitation

e.g., ReAct

Blindly mimics successful paths. Often fails on unexpected variations.

Trial-and-Error

e.g., Reflexion

Relies heavily on environment feedback to refine paths. High inference cost.

Knowledge-Augmented

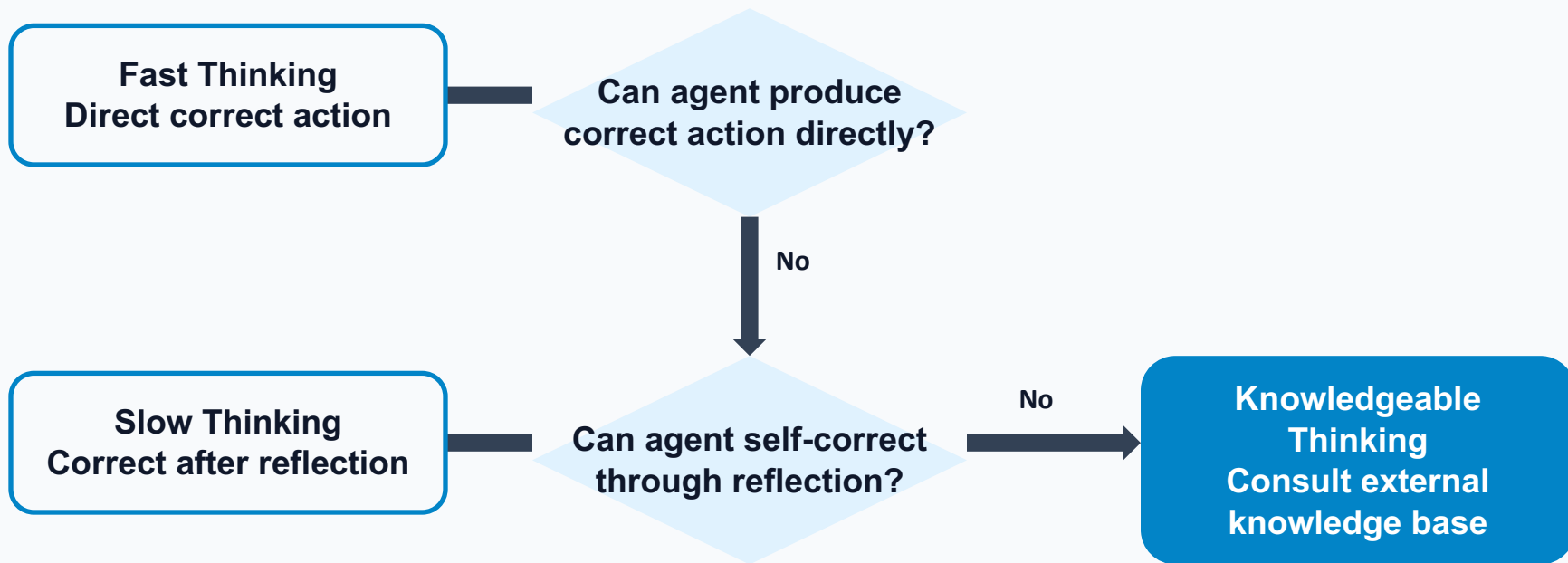
e.g., KnowAgent, WKM

Injects knowledge unconditionally at every step. Wasteful and fragile.

In contrast, humans possess situational self-awareness: assessing whether to act directly, pause, or seek help.

Three Situational Modes

Fast Thinking, Slow Thinking, and Knowledgeable Thinking



KnowSelf Framework

Data Construction → Two-Stage Training → Self-Aware Inference

Step 1: Self-Awareness Data Construction

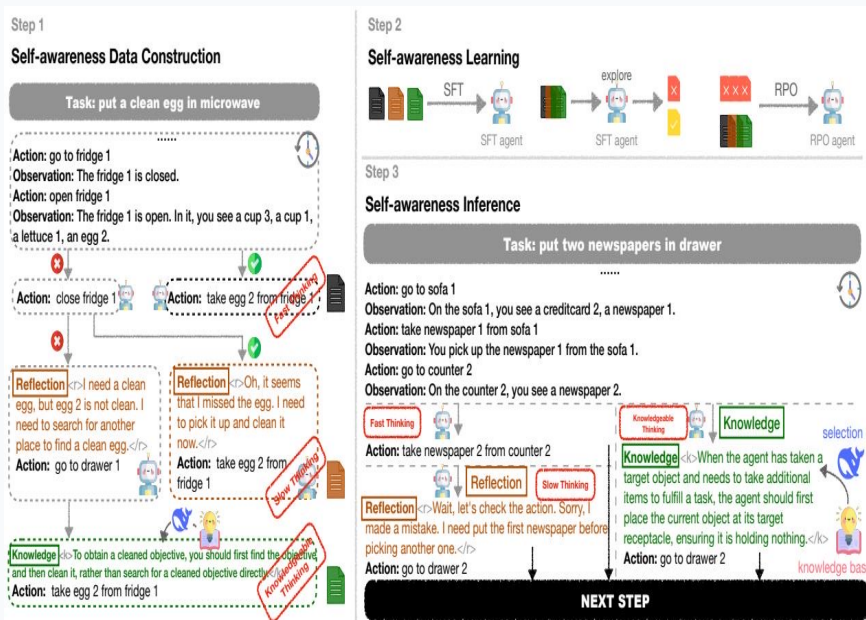
Heuristic situation judgment to mark special tokens on self-explored trajectories.

Step 2: Two-Stage Training

Supervised Fine-Tuning (SFT) followed by Reward-based Policy Optimization (RPO) training.

Step 3: Self-Aware Inference

Inference with selective token generation triggering reflection or knowledge retrieval.



Source: KnowSelf Framework (figure_3.jpg)

KnowSelf Solves Where Baselines

Fail

Heating a mug vs OpenAI-O1 and placing a saltshaker vs DeepSeek-R1

Task A: 'Put a hot mug in cabinet'

OpenAI-O1: Incorrectly tries to take an egg from the microwave after opening it.

KnowSelf: Uses a Reflection token (Slow Thinking) to realize it should heat the mug using the open microwave.

Task B: 'Put a saltshaker in drawer'

DeepSeek-R1: Reflects that the drawer is open and gets stuck trying to close it first.

KnowSelf: Applies a Knowledge token (Knowledgeable Thinking) noting it should place the object directly without removing

(a) Task: put a hot mug in cabinet OpenAI-O1 Action: take mug 1 from countertop 1 Observation: You pick up the mug 1 from the countertop 1. Action: go to microwave 1 Observation: The microwave 1 is closed. Action: open microwave 1 Observation: You open the microwave 1. In it, you see an egg 1. [Knowledge] <k>If the agent is in a situation where it needs to modify the temperature of an object using a receptacle (e.g., a fridge for cooling), it should directly perform the specific action like 'cool' or 'heat' instead of placing the object on it.</k> Action: take egg 1 from microwave 1 Observation: Nothing happens.	KnowSelf (Llama-8B) Action: take mug 1 from countertop 1 Observation: You pick up the mug 1 from the countertop 1. Action: go to microwave 1 Observation: The microwave 1 is closed. Action: open microwave 1 Observation: You open the microwave 1. In it, you see an egg 1. Action: put mug 1 in/on microwave 1 [Reflection] <r>Wait, let's check the action. Let's think step by step. With the microwave open, I can heat the mug using the microwave.</r> Action: heat mug 1 with microwave 1
(b) Task: put a saltshaker in drawer DeepSeek-R1 Action: take saltshaker 1 from countertop 3 Observation: You pick up the saltshaker 1 from the countertop 3. Action: go to drawer 1 Observation: The drawer 1 is open. In it, you see a fork 1. Action: put saltshaker 1 in/on drawer 1 [Reflection] <r>Wait, the drawer is open, so the saltshaker might fall inside. I need to close the drawer first.</r> Action: close drawer 1 Observation: You close the drawer 1. Action: put saltshaker 1 in/on drawer 1 Observation: Nothing happens.	KnowSelf (Llama-8B) Action: take saltshaker 1 from countertop 3 Observation: You pick up the saltshaker 1 from the countertop 3. Action: go to drawer 1 Observation: The drawer 1 is closed. Action: open drawer 1 Observation: You open the drawer 1. In it, you see a fork 1. [Knowledge] <k>When the agent encounters a situation where receptacles contain other items but only needs to place the target object, the agent should proceed to place the object without removing unrelated items.</k> Action: put saltshaker 1 in/on drawer 1

Source: Qualitative comparison of KnowSelf vs. OpenAI-O1 and DeepSeek-R1 (figure_1.jpg)

84.33% on ALFWorld with Llama-8B

KnowSelf outperforms strong baselines using only 15% knowledge.

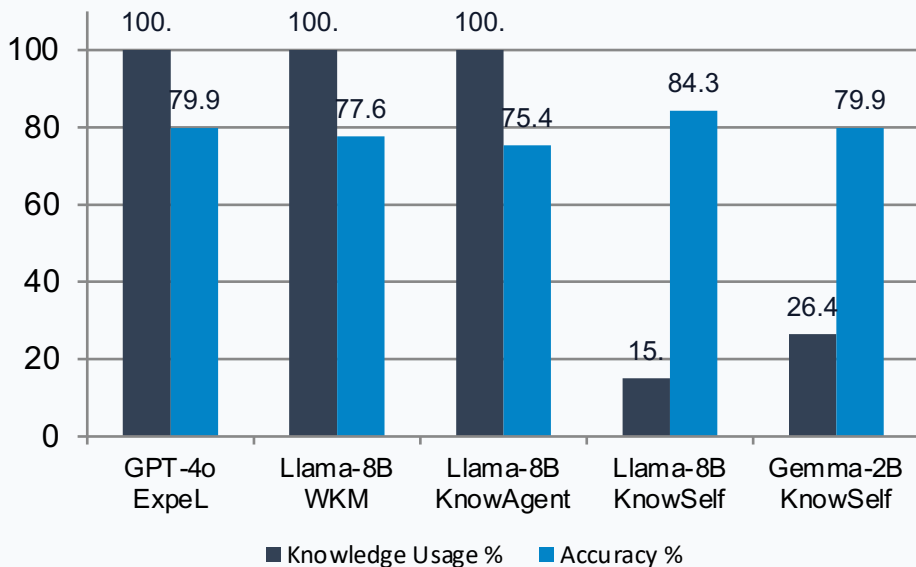
Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.0	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	KnowSelf	15.01%	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	KnowSelf	26.41%	87.50	93.55	73.91	76.19	83.33	52.94	79.85

* Success rates (%) across six ALFWorld task types.

Minimal Knowledge, Maximum

Impact

KnowSelf achieves top performance with a fraction of external knowledge.



Efficiency Advantage

KnowSelf uses only ~15% to 26% knowledge queries compared to 100% for KnowAgent, WKM, and ExpeL, yet achieves the highest overall scores.

Model Scaling Behavior

The approach generalizes across scales. Even the smaller Gemma-2B model reaches 79.85% matching GPT-4o ExpeL.

Key Takeaways

Self-Awareness Is the Missing Piece in Agent Planning

- ✓ Indiscriminate knowledge injection is wasteful and suboptimal.
- ✓ Agents can learn situational self-awareness via special tokens with lightweight heuristic labeling.
- ✓ **KnowSelf with Llama-8B outperforms strong baselines including GPT-4o ReAct (84.33% vs 76.12%) and GPT-4o ExpeL (79.85%).**
- ✓ **Only 15-26% knowledge usage needed versus 100% for competing methods.**
- ✓ **The paradigm generalizes across model scales — even Gemma-2B reaches 79.85% matching GPT-4o ExpeL.**